FEATURE-CONDITIONED AFFINE LIFTING OF QAOA PARAMETERS ACROSS CIRCUIT DEPTHS

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Abstract. Parameter interpolation across circuit depths ignores graph-dependent changes in the optimal schedule. This article studies a feature-conditioned affine lift from a depth-one reference to a depth-two schedule. The evaluation uses 144 ten-vertex neighborhoods extracted from six topology-unique, authenticated Stanford Gset graphs. Training and test feature signatures are disjoint in the primary six-fold protocol. The lift attains a mean depth-two relative regret of 0.774%, compared with 6.632% for interpolation and 4.050% for an affine map that omits graph features. Its normalized schedule mean-square error is 1.265×10^{-3}. A second source-held-out protocol reverses part of this ordering: the unconditioned affine map has lower regret than the conditioned map. Thus the evidence establishes improved transfer across unseen feature signatures, not universal source generalization. We give the regularized closed form and its arithmetic and storage costs, and prove a commutator bound that converts angle error into a MaxCut energy certificate. Every reported quantity is regenerated from authenticated graph files by a deterministic CPU replay.

Key words. QAOA, parameter transfer, MaxCut, affine regression, reproducibility

1. Depth lifting. For an unweighted graph $G = (V, E)$, let

$$C_G = \sum_{(u,v)\in E} \frac{I - Z_u Z_v}{2}, \qquad B = \sum_{v\in V} X_v. \tag{1.1}$$

The depth-p state alternates $e^{-i\gamma_j C_G}$ and $e^{-i\beta_j B}$, and its objective is $F_p(\boldsymbol{\gamma}, \boldsymbol{\beta}) = \langle C_G \rangle$ [2]. Interpolation provides a useful warm start when optimal schedules vary smoothly with depth [1, 4]. It does not represent changes associated with local density, degree dispersion, or short cycles.

The proposed lift receives the normalized depth-one reference $x_G = (\gamma_1/\pi, 2\beta_1/\pi)$ and the graph descriptor

$$\phi_G = (|E|, \bar{d}, s_d, t, w_3, w_4)/s, \tag{1.2}$$

where t is triangle count, $w_j = \operatorname{tr}(A^j)$, and s denotes fixed coordinate scales. With $z_G = (1, x_G, \phi_G)$ and normalized depth-two target $y_G \in \mathbb{R}^4$, the estimator is

$$\begin{aligned} \widehat{y}_G &= z_G \widehat{W}, \\ \widehat{W} &= \arg\min_W \{ \|ZW - Y\|_F^2 + \lambda \left\| D^{1/2} W \right\|_F^2 \}. \end{aligned} \tag{1.3}$$

The intercept is not penalized. The penalty is selected inside each training fold; no test labels enter model selection.

Theorem 1 (solution and resource bound)**.** *Let $Z \in \mathbb{R}^{N\times d}$, $Y \in \mathbb{R}^{N\times q}$, $D \succeq 0$, and suppose $H = Z^T Z + \lambda D$ is positive definite. Equation* (1.3) *has the unique solution*

$$\widehat{W} = H^{-1} Z^T Y. \tag{1.4}$$

Forming and solving the normal equations requires $O(Nd^2 + Ndq + d^3)$ arithmetic and $O(d^2 + dq)$ working storage beyond Z and Y. Prediction on M graphs requires $O(Mdq)$ arithmetic and $O(dq)$ model storage.

Proof. Expanding the objective gives

$$\begin{aligned} f(W) = {} & \operatorname{tr}(W^T Z^T Z W) - 2\operatorname{tr}(W^T Z^T Y) \\ & + \operatorname{tr}(Y^T Y) + \lambda \operatorname{tr}(W^T D W). \end{aligned} \tag{1.5}$$

Hence $\nabla f(W) = 2HW - 2Z^T Y$. Positive definiteness of H makes f strictly convex, so the unique stationary point is Eq. (1.4). Accumulating $Z^T Z$ and $Z^T Y$ costs $O(Nd^2 + Ndq)$ operations; a dense factorization of H costs $O(d^3)$. The stated storage and prediction counts follow from the dimensions of H, W, and each matrix product. □

Table 1: Authenticated Gset sources. Topology identifiers are SHA-256 prefixes after discarding weights; sources with the same identifier have identical unweighted topology. Only one member of each duplicate pair enters the audit.

Source	Vertices	Edges	Records	File SHA-256	Topology SHA-256	Role
G1	800	19176	24	`73bf704d8f`	`e1956ffd85`	analysis
G6	800	19176	0	`e3dd8b2920`	`e1956ffd85`	excluded duplicate topology
G11	800	1600	24	`c2a760d292`	`21bc870a7d`	analysis
G14	800	4694	24	`dc769b978a`	`e882f19f2a`	analysis
G18	800	4694	0	`6185af31fa`	`e882f19f2a`	excluded duplicate topology
G22	2000	19990	24	`9baeee06eb`	`ddfaa2b1cf`	analysis
G32	2000	4000	24	`9760fce6b6`	`4007cf1aea`	analysis
G43	1000	9990	24	`9af5445b4b`	`91ac033a6e`	analysis

2. Energy certificate. Schedule error is not an invariant measure of algorithmic quality. The next result supplies a graph-explicit conversion to energy error.

Theorem 2 (angle-to-energy stability)**.** *For unweighted MaxCut on n vertices and m edges, two depth-p schedules $\theta = (\boldsymbol{\gamma}, \boldsymbol{\beta})$ and $\theta' = (\boldsymbol{\gamma}', \boldsymbol{\beta}')$ satisfy*

$$|F_p(\theta) - F_p(\theta')| \le 2m^2 \left\|\boldsymbol{\gamma} - \boldsymbol{\gamma}'\right\|_1 + 2nm \left\|\boldsymbol{\beta} - \boldsymbol{\beta}'\right\|_1 . \tag{2.1}$$

If $F_p(\theta') > 0$, division by $F_p(\theta')$ gives the corresponding relative certificate.

Proof. For a circuit parameter t generated by a Hermitian matrix H_t, unitary invariance and the commutator inequality give

$$\begin{aligned} \left|\frac{\partial F_p}{\partial t}\right| &= |\langle\psi| i[H_t, C_G] |\psi\rangle| \\ &\le \|[H_t, C_G]\|_2 \le 2 \|H_t\|_2 \|C_G\|_2 . \end{aligned} \tag{2.2}$$

Because C_G is a sum of m projectors, $\|C_G\|_2 \le m$; because B is a sum of n commuting Pauli operators, $\|B\|_2 \le n$. Every cost-angle derivative is bounded by $2m^2$ and every mixer-angle derivative by $2nm$. Integrating the gradient along the segment joining θ and θ' proves Eq. (2.1). □

3. Authenticated experiment. The Stanford Gset files are verified against registered SHA-256 digests before parsing [3]. Duplicate unweighted topologies G6/G1 and G18/G14 are identified from sorted edge hashes and excluded. Twenty-four deterministic connected neighborhoods are drawn from each retained source. The resulting 144 records contain ten vertices; edge weights are discarded, so the estimand is unweighted MaxCut. Table 1 records the complete source audit.

Depth-one references use a fixed 13×9 grid. Depth-two labels use 25 local refinement evaluations and are therefore local references, not certified global optima. The primary folds hold out complete feature-signature classes, giving zero train–test signature overlap and zero exact-feature neighbors. The secondary protocol holds out one Gset source at a time. Both protocols rotate through six test folds. Schedule mean-square error and relative energy regret are reported separately.

4. Results. Table 2 compares only methods that receive the target graph's depth-one reference. On the collision-free signature protocol, graph conditioning reduces mean regret by 88.3% relative to interpolation and by 80.9% relative to the unconditioned affine map. The mean schedule error is 1.265×10^{-3}. On source-held-out folds, however, the unconditioned affine map has regret 1.350%, compared with 2.994% for graph conditioning. This reversal is a boundary of the result rather than an exception removed from the record.

Figure 1 resolves the aggregate values into protocol, fold, and certificate views. The signature-held-out gain persists in every fold except the fold in which both affine procedures attain zero measured regret. The commutator certificate has no violations among 144 predictions, but it is orders of magnitude looser than the observed gap. It is therefore suitable as a correctness envelope, not as a predictor-ranking surrogate.

The experiment does not establish optimal depth-two schedules, performance on weighted graphs, or transfer to graphs outside the six Gset topologies. It does establish a reproducible distinction between

Table 2: True depth-transfer controls, which receive each test graph's $p = 1$ reference. MSE covers the four normalized $p = 2$ coordinates.

Method	$p = 2$ MSE	R_2 (%)
INTERP (source)	6.622e-03	6.632
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$p = 1$ affine (source)	9.476e-04	1.350
$p = 1$ affine (signature)	2.165e-03	4.050
$p = 1$ + features affine (source)	4.468e-02	2.994
$p = 1$ + features affine (signature)	1.265e-03	0.774

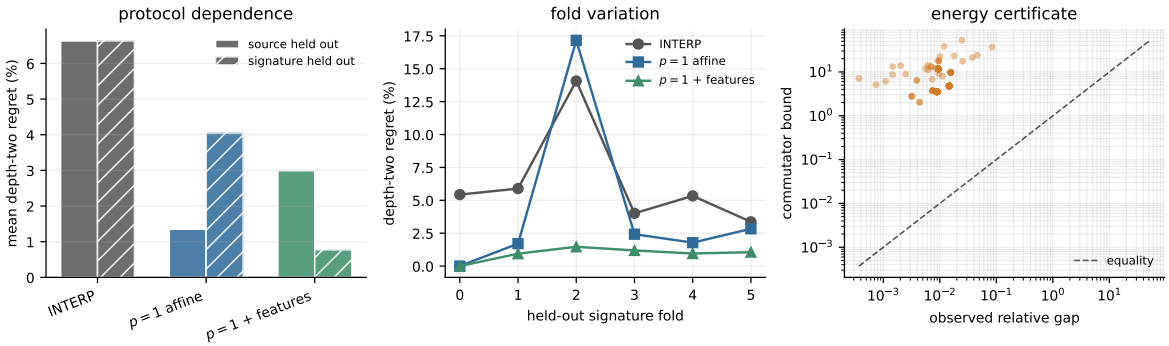


Fig. 1: Depth-transfer evidence. Left: mean depth-two regret under two nonoverlapping evaluations. Center: foldwise regret under the collision-free feature-signature protocol. Right: observed relative energy gap and the bound in Theorem 2; all points lie above the equality line because the ordinate is the certificate.

interpolation, schedule-only lifting, and graph-conditioned lifting under a split that prohibits identical feature signatures across training and test data.

5. Reproducibility. The package stores source addresses, byte hashes, topology hashes, fold assignments, hyperparameters, theorem checks, and all plotted values. A single command fetches the graph files, recomputes both protocols, regenerates the tables and figures, and compares the result with the locked record after excluding wall-clock fields. Tests independently check topology deduplication, fold disjointness, the affine construction, and all 144 energy certificates.

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